CAS Component Mathematics

Forty-two circuit parts, each reduced to one electrical verb, one cognitive verb, and one thing that happens to people. *Learn the verb pair and you have the part.*

READ THE BADGE BEFORE YOU TRUST THE MAPPING

ISO **Exact structural isomorphism.** Functional roles correspond one-for-one and the governing relation has the same form. Four parts earn this.

ENG **Engineered analog.** The role match is explicit, but a cognitive quantity, control input, or conservation rule is stipulated by the design. Fifteen parts.

PED **Pedagogical analogy.** The resemblance teaches. No quantitative relation transfers. One part, and it is fenced.

The badge grades *the mapping*, not the existence of the part. Every parameter here is a design choice — a setting you specify, like picking a 50-ohm resistor. None is a measurement of a person, an AI model, or a biological system. Logic and annotation entries carry a plate-level badge rather than an individual one.

S1 Flow and state retention

5



R1 Resistor

ENG

`limit / restrain activation`

> *The lad you spar with catches you square on the nose and grins. Everything you would throw back is there at once, exact, and what leaves your gloves over the rest of the round is a jab, and a jab, and a jab. At the bell your arms are shaking.*



C1 Capacitor

ENG

`retain / preserve composition`

> *Winter after winter your grandmother put her hands over yours on the needles until the stitches came even. She has been gone a long time. You pick up wool now and your hands find that same evenness before you have sat down — nothing recalled, no instruction left anywhere, only the hands.*

L1 Inductor

ENG

`oppose change / preserve an established pattern`

> *The man in the hire shop says the brake levers are the other way round on this bike. You know that. You repeat it to yourself, and for a mile you are fine. Then a van pulls out, your right hand does what it has always done, and the back wheel slides.*



CV1 Varactor

ENG

`tune capacitance / tune retention or resonance`

> *For two summers after you went over on the ankle, loose scree stopped you at the top of any slope. The physiotherapist worked on it for months, and for a while it felt worse. The stones are still exactly the stones. Now you go down them.*



Z1 LC tank

ENG

`resonate / favor one repeating rate`

> *There is one trot the mare gives you that you can sit to, and almost nothing either side of it. Faster and you are thrown about the saddle; slower and she drifts out of it. At that pace you go round the field for an hour, and neither of you is working.*

S2 Direction, threshold, and sensing

5



D1 Diode

ISO

`allow one way / enforce directed admissibility`

> *In the house you grew up in, nobody said what anything cost, and it is still the rule in yours. Your aunt asks twice what you paid for the piano, and twice you answer something else without deciding to. Telling her was never one of the things you could have done.*



ZD1 Zener diode

ENG

`trip and clamp / enforce a threshold with a ceiling`

> *Your cousin has had your drill a year, and the wallpaper table, and you have never asked for either back. Then he asks to put his boxes in your loft. Once, evenly: not the loft. He argues, and the same three words come back at the same volume, never louder. He keeps the drill.*



T1 Transformer

ENG

`convert level / translate across a licensed Domain seam`

> *The letter tells your father-in-law he must demonstrate continued eligibility by the date shown. He reads it three times and is none the wiser. What you tell him is: the council desk, his passport, the two forms they sent, all of it before the eighth. Next day he is there before it opens.*



PD1 Photodiode

ENG

`transduce light / admit external Idea through Mind intake`

> *Your landlord is in the middle of the room writing in his folder, and he says, without looking up, that he will be putting the flat up for sale. The pen keeps moving. You are standing exactly where you stopped, and your face is hot before you have thought one thing about it.*



BR1 Bridge rectifier

PED

`normalize scalar polarity / teaching only, on a named non-valence scalar`

> *After the coffee morning at the community centre, the tin is counted. Some weeks there is more in it than there should be, some weeks less. Either way the woman counting it says the same four words — we're three pounds out — and hunts just as long. Over or short never gets said.*

S3 Active and comparative parts

5



Q1 CJT

ENG

`control and amplify / let bias shape trigger into arrival`

> *Two of them hear the tutor say the same thing — that corner isn't square. The man has been making things since he was small; he reaches for the chisel. The woman beside him has gone quiet: a school art room thirty years back, a bowl held up, everyone looking.*



U1 Op-amp

ENG

`compare / amplify a commensurable difference`

> *Two stalls, the same tomatoes, tenpence a pound between them. By the end of the row he has counted up the summers he has been shopping at the dearer one, and what that comes to, and he is not going back. Tenpence. He has never asked the man what his prices are.*



Q2/Q3 Differential pair

ENG

`reject common mode / preserve a matched difference`

> *Every jar in the raspberry class is sweet, so by the third spoon the judge has stopped tasting for sweet. Two jars left, and one small thing between them: this one still tastes of fruit, that one tastes of the pan it was boiled in.*



Q4/Q5 Current mirror

ENG

`copy bias / share one comparison condition`

> *Two trays of the same seed, one bag of compost between them, the same windowsill, turned on the same evenings — he wants to know whether the new feed is worth buying. Move one tray nearer the stove for a week and whatever comes up tells him nothing about the feed.*

ΔI

Q6 Active load

ENG

`convert difference / land a branch difference as output`

> *The green and the grey have been over the back of the chair since Sunday. She has held each of them up against the light, put them down, walked past them, held them up again. Thursday morning she hears herself say it out loud — the green, it'll wear — and picks up the shears.*

S4 Rhythm and register conversion

5



OSC1 Oscillator

ENG

repeat / sustain a cycle

> *Nothing is playing it. The radio has been off since morning, nobody in the house is humming, and the same scrap of a chorus keeps coming round, its last word handing straight back to its first. It was going at noon over the ironing. It is going now, while she puts the lights out.*



Y1 Crystal

ENG

stabilize frequency / hold a low-drift cadence

> *He will not hold the boat. The vicar has asked, the farmer has asked, and a family once came down the slip waving both arms and watched it go without them. High water or low, June or the January dark, the rope comes off the post at the same minute.*

PLL

U2 Phase-locked loop

ISO

lock / entrain an internal rhythm

> *He joins the back row of the choir in October and spends the first night a beat behind everyone. By the third week he breathes where the row breathes and comes in where they come in. Singing it alone on the way home, he takes it at their speed, not his own.*

A/D

U3 ADC

ISO

quantize / continuous register becomes discrete

> *Nobody alive sings it but him, so she is putting it onto paper in pencil, bar by bar. Now it can be sung by people who never heard him. The lean he puts on the third line, and how long he holds it before letting go, is not on the paper.*

D/A

U4 DAC

ISO

reconstruct / discrete register becomes continuous

> *Four hours on the train north, with three things to say and the order fixed. Once he is there, none of it comes out in threes. He carries the bags in, stays until Sunday, and the second thing turns out to be the way he sits at the table and lets his uncle finish.*

S5 Logic and routing

11 · plate-level ENG



21 AND

–

require all / detect all-active prerequisites or conflict

> *The parking permit wants all three on the same morning: the photograph, the signed declaration, and a bill dated inside the last three months. Twice it has come back, and twice only the bill was out of date. Perfect photograph, perfect signature, no permit.*



22 OR

`require any / accept one or more thresholded flags`

> *Any one of the three closes the pool: thunder overhead, a lifeguard off sick, or the water gone cloudy. Today it is only the water — the sun is out, the staff are all in — and the notice is already up at the turnstile.*



23 NOT

`complement / negate one explicit firing flag`

> *The baker turns the sign in the door from OPEN to CLOSED at two. The ovens are still warm, she is still working, and six loaves sit cooling on the rack behind her. The handle simply stops turning for anyone out on the pavement.*



24 NAND

`reject all-active / invert an AND result`

> *Both open at once is the one thing the crossing will never do. The barrier lifts only after the lights have gone dark; the lights only clear once the barrier is down. Some minutes nothing moves at all — the lorry waits, no train in sight — and that suits the crossing perfectly.*



25 NOR

`require none / accept only when every flag is inactive`

> *The lift will not move while anything is in its way: a suitcase against the beam, the inner gate not quite shut, the outer gate left open on the fourth floor. Clear all three and it goes without a sound. Leave one and it sits there humming, going nowhere.*



26 XOR

`distinguish / accept exactly one active flag`

> *One answer per line on the quiz slip. The table hedges and writes both Mercury and Venus in the box; the marker gives them nothing, exactly as if the line had been left blank. The table behind them wrote one word, wrong or right, and got it marked.*



27 XNOR

`agree / detect equal flags, including both-inactive`

> *Both squares blank on the shared calendar and nobody rings. Both filled in with the same trip to the coast and nobody rings either — your sister sees it, you see it, there is nothing to arrange. It is the Saturday blank on her side and full on yours that starts the phoning.*



28 MUX

`choose / route one selected input to one output`

> *Forty things sit behind the glass; the buttons decide which one falls. B4 sends down the crisps, B5 the chocolate, and both arrive in the same tray at the bottom. Punch nothing and nothing comes — the machine just stands on the platform with all forty still in their rows.*

DMX

29 DEMUX

–

`dispatch / send one input to one selected output`

> *One bagful, one skip. The man in the hi-vis looks in, says "wood", and points to the far end. Not wood and garden. Not try both and see. Whatever else is in that bag goes over the side with the wood, or you stand in the cold and pull it out first.*



30 Relay

–

`isolate and switch / let Cause control a separate Effect path`

> *The dial in the hall is the size of a saucer and turns with one finger. At six it clicks, and two floors down the boiler thumps into life; within minutes every radiator in the building is ticking. Nothing hot ever reaches the hall. The dial stays cold in your hand.*



31 Schmitt trigger

–

`commit and hold / prevent threshold chatter`

> *Rain, then no rain, then rain: the covers have been on and off the square four times since lunch. At four the umpire walks out, looks up once, and calls it. The sky clears at half past and stays clear. The covers stay on and everybody goes home.*

S6 Regulation, protection, annotation

11 · not individually graded

REG

32 Voltage regulator

–

`restore / maintain a setpoint`

> *You never notice it happening and you never do anything about it: snapped at over breakfast, praised mid-morning, a long post-office queue at four — and the same list of chores gets worked through all day anyway. By evening, chopping onions, you sound like yourself again. Not better than usual. Yourself.*



33 Fuse

–

`sever once / stop an unsafe path for the stated lifetime`

> *You handed the keys across the counter yourself and lifted your own mug off the hook, and by Wednesday another name was on the rota. Months earlier, alone on the bus, you had told yourself: dock my wages once more and I go that day. They docked them in March.*



34 Circuit breaker

–

`trip and reset / stop a recoverable fault path`

> *Not for the first time this term, you pull into the lay-by and cut the engine mid-shriek. Nobody moves until the noise drains out of the car; then you start up and finish the school run. The children go quiet now the moment the car slows.*



35 Drive rail

`supply / make motive potential available`

> *Everything else went fine today — the washing is out, the shelf is fixed, the bathroom scrubbed. Then you sit down with the book you chose yourself, wanting to read it, and nothing in it catches. On Tuesday a colleague asks you something you cannot answer; that night the same book keeps you reading until dawn.*



36 Reference / rest node

`define zero / set an engineered baseline`

> *You could have gone on judging every night against your own bed at home. Instead, on the first morning of the camping trip, you decide that a cold tent at first light is what comfortable will mean for the fortnight. Halfway through, the drizzle is nothing and tea from a flask tastes extraordinary.*



37 Probe / test point

`observe / inspect one named point without becoming its signal`

> *From your front room you can see that the young defender has stopped turning to his left. You do not know what he has done to himself. You cannot get word to him. He plays on, and the match goes on.*



38 Switch / enable

`connect / enable or inhibit one path`

> *Tonight you shut the bedroom door on the cat, instead of leaving it open a hand's width for her the way you always do. She scratches at it, then settles on the landing. You lie there listening, ready to get up and let her in if she starts again.*



39 Meter

`measure / report one named quantity by a named method`

> *The plumber says the job is two days. You ask what is inside the two days: two days on site with the water off, or door to door, counting the journey each way and the morning she spends waiting for a part. She means the second. You had assumed the first.*



40 Net label

`name / type the whole state or scalar`

> *You write the week out on the back of an envelope: the sharp reply to the neighbour, the email you have still not answered, the second night of bad wine, the run you skipped twice. Then you write one word underneath — brother — and every line above it becomes the same line.*



41 Solid forward path

`advance / show the main forward path`

> *You put off the dentist in April because the ache faded on its own. In August, halfway through a slice of toast, the tooth splits. By Tuesday morning half your face is numb, a stranger's hands are in your mouth, and you are booked back in for a crown on Friday.*

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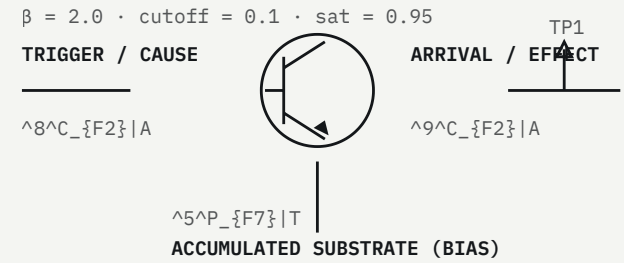
42 Dashed feedback path

`return / show a sampled feedback path`

> *Each Monday you say less in the meeting than you did the Monday before. You went quiet after your idea was talked over; your manager read the quiet as not minding much and stopped asking you first; and her not asking is the reason you give yourself, in the corridor, for staying quiet again.*

ATOM The one circuit to know by heart

Circuit 1



A trigger enters, accumulated composition biases it, and a *different* Idea arrives. This is a trigger-to-arrival transition — not a synonym for reasoning. Terminal roles are the CAS abstraction, not claims about a transistor's collector, base and emitter.

S7 The other nine, in one line each

Circuits 2–10

02 Directed domain descent

`P → C → E → O` by directed admissibility; `O → P` is a labelled feedback path, never an upward override

03 Gain cascade

three CJTs, each with its own bias · 1.5 × 1.0 × 2.0 = 3.0 · still bounded by sat = 0.8

04 RC-compensated feedback

outcome through R_fb and C_fb (τ = 5 min) returns only a smoothed trend, at β_fb = 0.2

05 Ambiguity stabilizer

^2 flag AND ^3 flag = conflict · Schmitt on 0.7 / off 0.3 holds the decision while it wobbles

06 CSB red path

refusal sense 0.05 sits below the red clamp 0.1, so the fuse is reachable · it cuts the green answer, never the refusal

07 Differential comparison

two commensurable activations, one shared bias · the difference lands as ^6 decomposition + ^9 arrival

08 Rhythm and entrainment

`LC tank selects the resonant rate (^4 Vibration) · PLL locks to an external beat (^7 Rhythm) · crystal is a reference, not a Noetic`

09 Retention cell

`C = 200, leakage = 0.001, refresh pump · a low charge alone cannot tell never-written from leaked-away`

10 Drive supply

`7 rails · dials 3–9, involution divergence ≤ 4 · O enables only after P, C, E are power-good · brownout trip 4.2 / release 5.4 · emergency fails closed to silence`

OPS Three operators, and what they promise

v7.7

+	**Co-presence.** Manifestations sit together and stay distinct. Say “associates with” and check it still reads true.
×	**Integration.** Licensed only with a named emergent result. Say “intensifies through”. Same-Noetic multi-register `×` without a named result is invalid.
|⟨…⟩	**Register occupancy.** States where something lives. Makes no claim at all about interaction.

GUARD Six ways this goes wrong

check before you classify

1. **^3 Negative is repulsion. ^6 Male is decomposition.** Auditing or breaking down an argument is ^6 — never ^3, however critical it feels.
2. **^8 Above/Cause is associative triggering, not reasoning.** It brings a *different* Idea to Mind. The thing that arrives is ^9 Below/Effect.
3. **^5 Female is composition, not a container.** The Mind *is* what it accumulated. It is not a box holding experiences.
4. **A part is a role, not a Noetic.** One part may act on Noetic, Domain, Drive, Register, or a board-level control quantity. Never force one component onto one Noetic.
5. **Reference is an engineered zero.** Calling a cognitive rest state ^0 Idea is a convention you chose, not an identity.
6. **AND detects both-active. XOR is exactly-one. XNOR includes both-inactive.** A both-active conflict detector is AND — reaching for XOR here is the common slip.

What this is. A carry-sized companion to the CAS PCBA Engineer material — the 42 frozen component and annotation entries, each with its verb pair, its mapping grade, and one everyday example. It is a teaching and design reference, not evidence.

What the numbers are. Every parameter shown — β, τ, thresholds, dial ranges — is a design choice you would specify in a schematic. None is a measurement of a person, an AI model, or a biological system, and none is a benchmark result.

Domain colours. **P** Principled · **C** Cognitive · **E** Evaluative · **O** Operational. Colour reinforces; the labels and shapes carry the claim, so every plate survives greyscale printing.

Source. Verb pairs, grades and circuit parameters are taken from the Complete Memory Deck and plates S1–S6 of the CAS Component Companion. The everyday examples are new here, and were drift-checked against the six guardrails above.